Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

The wavelength of H_{α} line of Balmer series is $X\ \mathring{A}$. What is the wavelength of H_{β} line of Balmer series.

(A) $\mathrm{X}\frac{108}{80}\mathring{A}$

(B) $\mathrm{X}\frac{80}{108}\mathring{A}$

(C) $\frac{1}{\mathrm{X}}\frac{80}{108}\mathring{A}$

(D) $\frac{1}{\mathrm{X}}\frac{108}{80}\mathring{A}$

Q.2

A dye absorbs a photon of wavelength λ and reemits the same energy into two photons of wavelength λ_1 and λ_2 respectively. The wavelength λ is related with λ_1 and λ_2 as:

(A) $\lambda = \frac{\lambda_1+\lambda_2}{\lambda_1\lambda_2}$

(B) $\lambda = \frac{\lambda_1\lambda_2}{\lambda_1+\lambda_2}$

(C) $\lambda = \frac{\lambda_1^2\lambda_2^2}{\lambda_1+\lambda_2}$

(D) $\lambda = \frac{\lambda_1\lambda_2}{(\lambda_1+\lambda_2)^2}$

Q.3

The wave number of the first emission line in the Balmer series of H -Spectrum is:
$(\mathrm{R} = \text{Rydberg constant})$:

(A) $\frac{5}{36}R$

(B) $\frac{9}{400}R$

(C) $\frac{7}{6}R$

(D) $\frac{3}{4}k$

Q.4

What is the maximum wavelength line in the Lyman series of He^{+} ion?

(A) $3\mathrm{R}$

(B) $\frac{4}{4\mathrm{R}}$

(C) $\frac{1}{3R}$

@IITJEE_Advanced

(D) None of these

Q.5

Find the value of wave number $(\overline{\mathrm{v}})$ in terms of Rydberg's constant, when transition of electron takes place between two levels of He^{+} ion whose sum is 4 and difference is 2 .

(A) $\frac{8 \mathrm{R}}{9}$

(B) $\frac{32 \mathrm{R}}{9}$

(C) $\frac{3 \mathrm{R}}{9}$

(D) None of these

Q.6

The electron energy in hydrogen atom is given by $\mathrm{E}=\left(-21.7 \times 10^{-12}\right) / \mathrm{n}^{2}$ ergs. By calculating the energy required to remove an electron completely from the $\mathrm{n}=2$ orbit, find out the longest wavelength (in cm) of light that can be used to cause this transition?

(A) 3.67×10^{-5} cm

(B) 2.67×10^{-5} cm

(C) 4.67×10^{-5} cm

(D) 0.67×10^{-5} cm

Q.7

The λ of H_{α} line of Balmer series is $6500 \mathring{A}$. What is the λ of H_{β} line of Balmer series?

(A) $4814.8 \mathring{A}$

(B) $5814.8 \mathring{A}$

(C) $2867.4 \mathring{A}$

(D) $9395.3 \mathring{A}$

Q.8

A near UV photon of $300 \mathrm{~nm}$ is absorbed by a gas and then re-emitted as two photons. One photon is red with wavelength $760 \mathrm{~nm}$. Hence, wavelength of the second photon is

(A) $460 \mathrm{~nm}$

(B) $1060 \mathrm{~nm}$

(C) $496 \mathrm{~nm}$

(D) $300 \mathrm{~nm}$

@IITJEE_Advanced

Q.9

To move the electron in one H atom from the ground state to the second excited state, $12.084\mathrm{eV}$ are needed. How much energy is needed to cause $1\ \mathrm{mol}$ of H atoms to undergo this transition ?

(A) $728\ \mathrm{kJ}$

(B) $984\ \mathrm{kJ}$

(C) $1036\ \mathrm{kJ}$

(D) $1164\ \mathrm{kJ}$

Q.10

If the shortest wavelength of H atom in Lyman series is x, then longest wavelength in Balmer series of He^{+} is

(A) $\frac{9\mathrm{x}}{5}$

(B) $\frac{36\mathrm{x}}{5}$

(C) $\frac{x}{4}$

(D) $\frac{5x}{9}$

Q.11

Which of the following radiations has the highest wave number?

(A) X-rays

(B) Microwaves

(C) I.R. rays

(D) Radio waves

Q.12

The frequency of light emitted for the transition $n = 4$ to $n = 2$ of the He^{+} is equal to the transition in H atom corresponding to which of the following ?

(A) $n = 2$ to $n = 1$

(B) $n = 3$ to $n = 2$

(C) $n = 4$ to $n = 3$

(D) $n = 3$ to $n = 1$

Q.13

In hydrogen atomic spectrum, a series limit is found at $12186.3\ \mathrm{cm}^{-1}$. Then it belong to

@IITJEE_Advanced

(A) Lyman series

(B) Balmer series

(C) Paschen series

(D) Brackett series

Q.14

The limiting line in Balmer series will have a frequency of (Rydberg constant, $R_{\infty} = 3.29 \times 10^{15}$ cycles $/\mathrm{s}$)

(A) $8.22 \times 10^{14}\ \mathrm{s}^{-1}$

(B) $3.29 \times 10^{15}\ \mathrm{s}^{-1}$

(C) $3.65 \times 10^{14}\ \mathrm{s}^{-1}$

(D) $5.26 \times 10^{13}\ \mathrm{s}^{-1}$

Q.15

Which of the following statements is not true?

(A) Lyman spectral series of hydrogen atom lies in the ultraviolet region of electromagnetic radiation

(B) Balmer spectral series of hydrogen atom lies in the visible region of electromagnetic radiation

(C) Pashen spectral series of hydrogen atom lies in the visible region of electromagnetic radiation

(D) Brackett spectral series of hydrogen atom lies in the infrared region of electromagnetic radiation

Answers & Solutions

Q.1 Answer:

$\mathrm{X}\frac{80}{108}\mathring{A}$

Solution:

H_{α} **line of Balmer series means first line of Balmer series.** $n_1 = 2, n_2 = 3$

$\bar{v} = \frac{1}{\lambda_{\alpha}} = \mathrm{R}\left(\frac{1}{2^2} - \frac{1}{3^2}\right) = \frac{5\mathrm{R}}{36}$

$\therefore \lambda_{\alpha} = \frac{36}{5\mathrm{R}} = \mathrm{X}$

H_{β} **line of Balmer series means, second line of Balmer series,**

$\mathrm{n}_1 = 2, \mathrm{n}_2 = 4\ \bar{v} = \frac{1}{\lambda_{\beta}} = \mathrm{R}\left(\frac{1}{2^2} - \frac{1}{4^2}\right) = \frac{3\mathrm{R}}{16}$

$\therefore \lambda_{\beta} = \frac{16}{3\mathrm{R}} = \mathrm{X}$

when $\frac{36}{3R} = X$

Then $\frac{16}{3R} = \frac{X \times 5R \times 16}{36 \times 3R} = \frac{80X}{108}\mathring{A}$

Q.2 Answer:

$\lambda = \frac{\lambda_1 \lambda_2}{\lambda_1 + \lambda_2}$

@IITJEE_Advanced

Solution:

$$E = E_1 + E_2; \quad \frac{hc}{\lambda} = \frac{hc}{\lambda_1} + \frac{hc}{\lambda_2}$$
$$\Rightarrow \frac{hc}{\lambda} = hc\left(\frac{\lambda_2+\lambda_1}{\lambda_1\lambda_2}\right); \quad \lambda = \frac{\lambda_1\lambda_2}{\lambda_1+\lambda_2}$$

Q.3 Answer:

$\frac{5}{36}R$

Solution:

$$\bar{v} = RZ^2\left(\frac{1}{2^2} - \frac{1}{3^2}\right)$$
$$= R\left(\frac{1}{4} - \frac{1}{9}\right) = \frac{5R}{36}$$

Q.4 Answer:

$\frac{1}{3R}$

Solution:

$$\frac{1}{\lambda} = RZ^2\left[\frac{1}{n_1^2} - \frac{1}{n_2^2}\right] = R \times 2^2\left[\frac{1}{1^2} - \frac{1}{2^2}\right].$$
$$\Rightarrow 3R; \lambda = \frac{1}{3R}$$

Q.5 Answer:

$\frac{32R}{9}$

Solution:

$$n_1 + n_2 = 4; n_2 - n_1 = 2; \therefore n_1 = 1, n_2 = 3$$
$$\overline{v} = R(2)^2\left[\frac{1}{1^2} - \frac{1}{3^2}\right] = \frac{32R}{9}$$

Q.6 Answer:

3.67×10^{-5} cm

Solution:

$$\Delta E = E_\infty - E_2 = [0] - \left[-\frac{21.7\times10^{-12}}{4}\right]$$
Since $E_n = -\frac{21.7\times10^{-12}}{n^2}$ ergs $= \frac{21.7\times10^{-12}}{4}$ $\Delta E = hv$
$\Delta E = \frac{hc}{\lambda}$ $\frac{21.7\times10^{-12}}{4} = 6.627 \times 10^{-27} \times \frac{3\times10^{10}}{\lambda}$
whence $\lambda = 3.67 \times 10^{-5}$ cm

Q.7 Answer:

$4814.8\mathring{A}$

Solution:

@IITJEE_Advanced

For H_α lines of Balmer series $n_1 = 2, n_2 = 3$

For H_β line of Balmer series $n_1 = 2, n_2 = 4$

$\therefore \quad \frac{1}{\lambda_{H_\alpha}} = R_H \left[\frac{1}{2^2} - \frac{1}{3^2}\right]$ ...(1)

and $\frac{1}{\lambda_{H\beta}} = R_H \left[\frac{1}{2^2} - \frac{1}{4^2}\right]$ ...(2)

By Eqs. (1) and (2)

$\frac{\lambda_\beta}{\lambda_\alpha} = \frac{\frac{1}{4} - \frac{1}{9}}{\frac{1}{4} - \frac{1}{16}} \quad \therefore \quad \lambda_\beta = \lambda_\alpha \times \left[\frac{80}{108}\right] = 6500 \times \frac{80}{108} = 4814.8\mathring{A}$

Q.8 Answer:

496 nm

Solution:

Energy values are additives

$E = E_1 + E_2$

$\frac{hc}{\lambda} = \frac{hc}{\lambda_1} + \frac{hc}{\lambda_2}$

$\frac{1}{\lambda} = \frac{1}{\lambda_1} + \frac{1}{\lambda_2}$

$\lambda = 300 \text{ nm}, \lambda_1 = 760 \text{ nm}$

$\therefore \lambda_2 = 495.6 \text{ nm}$

$\approx 496 \text{ nm}$

Q.9 Answer:

1164 kJ

Solution:

$$\begin{aligned} \Delta E &= 12.084 \text{eV} \\ &= 12.084 \times 1.6 \times 10^{-19} \text{ J per atom} \\ &= 12.084 \times 1.6 \times 10^{-19} \times 6.02 \times 10^{23} \text{ J mol}^{-} \\ &= 1164 \text{ kJ} \end{aligned}$$

Q.10 Answer:

$\frac{9x}{5}$

Solution:

$\frac{1}{\lambda} = \overline{R}_H Z^2 \left[\frac{1}{n_1^2} - \frac{1}{n_2^2}\right]$

For Lyman series $n_1 = 1$,

λ is shortest if $n_2 = \infty$

$\therefore \overline{R}_H = \frac{1}{x}$ for H -atom $(Z = 1)$

For Balmer series, $n_1 = 2$

λ is longest if $n_2 = 3$

Here $\frac{1}{\lambda_{max}} = \left(\frac{1}{x}\right)(2)^2 \left[\frac{1}{2^2} - \frac{1}{3^2}\right]$ for He^+ ion

@IITJEE_Advanced

$(Z = 2)$
$\therefore \lambda_{max} = \frac{9x}{5}$

Q.11 Answer:

X-rays

Solution:

Since $v \propto \bar{v} \propto \frac{1}{\lambda}$, the order is X-Rays, I.R. Rays, Microwaves, Radiowaves

Q.12 Answer:

$n = 2$ to $n = 1$

Solution:

For He, $\bar{v} = \frac{1}{\lambda} = R_H Z^2 \left(\frac{1}{2^2} - \frac{1}{4^2}\right)$
$= R_H (2)^2 \left(\frac{1}{2^2} - \frac{1}{4^2}\right) = R_H \left(\frac{1}{(1)^2} - \frac{1}{(2)^2}\right)$
For H, $\bar{v} = \frac{1}{\lambda} = R_H \left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)$
For same frequency, $R_H \left(\frac{1}{(1)^2} - \frac{1}{(2)^2}\right) = R_H \left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)$
$\therefore \quad \frac{1}{n_1^2} - \frac{1}{n_2^2} = \frac{1}{1^2} - \frac{1}{2^2}$
$\therefore \quad n_1 = 1 \& n_2 = 2$

Q.13 Answer:

Paschen series

Solution:

Series limit is the last line of the series, i.e. $n_2 = \infty . \therefore \bar{v} = \frac{1}{\lambda} = \mathrm{R}\left[\frac{1}{\mathrm{n}_1^2} - \frac{1}{\mathrm{n}_2^2}\right] =$
$\mathrm{R}\left[\frac{1}{\mathrm{n}_1^2} - \frac{1}{\infty^2}\right] = \frac{\mathrm{R}}{\mathrm{n}_1^2}$
$\because \overline{\mathrm{v}} = 12186.3 = \frac{109677.76}{\mathrm{n}_1^2}$
$\Rightarrow n_1^2 = \frac{109677.76}{12186.3} = 9 \Rightarrow n_1 = 3$
- The line belongs to Paschen series.

Q.14 Answer:

$8.22 \times 10^{14}\ \mathrm{s}^{-1}$

Solution:

$\bar{v} = \frac{1}{\lambda} = R_H Z \left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)$
In Balmer series $n_1 = 2 \& n_2 = 3, 4, 5 \ldots$ **. Last line of the spectrum is called series limit. Limiting line is the line of shortest wavelength and high energy when** $n_2 = \infty$
$\therefore \quad \overline{\mathrm{v}} = \frac{1}{\lambda} = \frac{R_H}{n_1^2} = \frac{3.29 \times 10^{15}}{2^2} = \frac{3.29 \times 10^{15}}{4}$
$= 8.22 \times 10^{14}\ \mathrm{s}^{-1}$

@IITJEE_Advanced

Q.15 Answer:

Pashen spectral series of hydrogen atom lies in the visible region of electromagnetic radiation

Solution:

Paschen spectral series lies in the near infrared region of electromagnetic radiation.

@IITJEE_Advanced